MSG No. 133.469 P_C4_2/nbc [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `m'm'm'.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,4,5,6,25,28,29,30]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,3,7,8,26,27,31,32]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[9,12,13,14,17,20,21,22]`
4	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[10,11,15,16,18,19,23,24]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4m'2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,6,10,11,23,24,28,29]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,9,14,20,21,31,32]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[4,5,15,16,18,19,25,30]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,8,12,13,17,22,26,27]`

Table 3: Wyckoff site: `4c`, site symmetry: `m'm'm'.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[1,4,5,6,25,28,29,30]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[2,3,7,8,26,27,31,32]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[9,12,13,14,17,20,21,22]`
4	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[10,11,15,16,18,19,23,24]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4'm'2`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,6,7,8,26,27,28,29]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,4,5,25,30,31,32]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[9,14,15,16,18,19,20,21]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[10,11,12,13,17,22,23,24]`

Table 5: Wyckoff site: `8e`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[1,7,22,24]`
2	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,5,19,20]`
3	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,4,18,21]`
4	$\left[0, 0, \frac{3}{4}\right]$	`[6,8,17,23]`
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[9,15,30,32]`
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,13,27,28]`
7	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[11,12,26,29]`

8	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[14,16,25,31]`

Table 6: Wyckoff site: `8f`, site symmetry: `2'/m'.`.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,9,22,30]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,10,19,27]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,11,18,26]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,12,21,29]`
5	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,13,20,28]`
6	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[6,14,17,25]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[7,15,24,32]`
8	$[0,\ 0,\ 0]$	`[8,16,23,31]`

Table 7: Wyckoff site: `8g`, site symmetry: `2m'm'.`.

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,6,28,29]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,3,31,32]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[4,5,25,30]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[7,8,26,27]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[9,14,20,21]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[10,11,23,24]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[12,13,17,22]`
8	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[15,16,18,19]`

Table 8: Wyckoff site: `8h`, site symmetry: `2m'm'.`.

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[1,6,28,29]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,3,31,32]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[4,5,25,30]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[7,8,26,27]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[9,14,20,21]`
6	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[10,11,23,24]`
7	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[12,13,17,22]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[15,16,18,19]`

Table 9: Wyckoff site: `8i`, site symmetry: `m'm'2.`.

No.	position	mapping
1	$\left[\frac{3}{4},\ y,\ \frac{1}{2}\right]$	`[1,5,28,30]`
2	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ 0\right]$	`[2,8,27,31]`
3	$\left[y,\ \frac{3}{4},\ 0\right]$	`[3,7,26,32]`

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[4,6,25,29]
5	$\left[\frac{1}{4},\ -y,\ \frac{1}{2}\right]$	[9,13,20,22]
6	$\left[y+\frac{1}{2},\ \frac{1}{4},\ 0\right]$	[10,16,19,23]
7	$\left[-y,\ \frac{1}{4},\ 0\right]$	[11,15,18,24]
8	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[12,14,17,21]

Table 10: Wyckoff site: 8j, site symmetry: m'm'2.

No.	position	mapping
1	$\left[\frac{3}{4},\ y,\ 0\right]$	[1,5,28,30]
2	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ \frac{1}{2}\right]$	[2,8,27,31]
3	$\left[y,\ \frac{3}{4},\ \frac{1}{2}\right]$	[3,7,26,32]
4	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ 0\right]$	[4,6,25,29]
5	$\left[\frac{1}{4},\ -y,\ 0\right]$	[9,13,20,22]
6	$\left[y+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	[10,16,19,23]
7	$\left[-y,\ \frac{1}{4},\ \frac{1}{2}\right]$	[11,15,18,24]
8	$\left[\frac{1}{4},\ y+\frac{1}{2},\ 0\right]$	[12,14,17,21]

Table 11: Wyckoff site: 16k, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[1,22]
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[2,19]
3	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[3,18]
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	[4,21]
5	$\left[0,\ \frac{1}{2},\ -z\right]$	[5,20]
6	$[0,\ 0,\ z]$	[6,17]
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[7,24]
8	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[8,23]
9	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[9,30]
10	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[10,27]
11	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[11,26]
12	$\left[\frac{1}{2},\ 0,\ z\right]$	[12,29]
13	$\left[0,\ \frac{1}{2},\ z\right]$	[13,28]
14	$[0,\ 0,\ -z]$	[14,25]
15	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,32]
16	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[16,31]

Table 12: Wyckoff site: 16l, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[1,23]
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	[2,21]

continued ...

Table 12

No.	position	mapping
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[3,20]
4	$\left[x,\ -x,\ \frac{1}{4}\right]$	[4,19]
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[5,18]
6	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	[6,24]
7	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[7,17]
8	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[8,22]
9	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[9,31]
10	$\left[x,\ -x,\ \frac{3}{4}\right]$	[10,29]
11	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[11,28]
12	$\left[-x,\ x,\ \frac{3}{4}\right]$	[12,27]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[13,26]
14	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[14,32]
15	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	[15,25]
16	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[16,30]

Table 13: Wyckoff site: 16m, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ x,\ \frac{3}{4}\right]$	[1,7]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[2,5]
3	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[3,4]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[6,8]
5	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[9,15]
6	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[10,13]
7	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[11,12]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[14,16]
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[17,23]
10	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[18,21]
11	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[19,20]
12	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[22,24]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[25,31]
14	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[26,29]
15	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[27,28]
16	$\left[x,\ x,\ \frac{1}{4}\right]$	[30,32]

Table 14: Wyckoff site: 16n, site symmetry: m'..

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	[1,30]
2	$\left[\frac{1}{2}-y,\ x,\ 0\right]$	[2,27]
3	$\left[y,\ \frac{1}{2}-x,\ 0\right]$	[3,26]
4	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[4,29]
5	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}\right]$	[5,28]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[6,25]

continued ...

Table 14

No.	position	mapping
7	$[y,\ x,\ 0]$	`[7,32]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ 0\right]$	`[8,31]`
9	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[9,22]`
10	$\left[y+\frac{1}{2},\ -x,\ 0\right]$	`[10,19]`
11	$\left[-y,\ x+\frac{1}{2},\ 0\right]$	`[11,18]`
12	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[12,21]`
13	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	`[13,20]`
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[14,17]`
15	$[-y,\ -x,\ 0]$	`[15,24]`
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[16,23]`

Table 15: Wyckoff site: `16o`, site symmetry: `.m'.`

No.	position	mapping
1	$\left[\frac{3}{4},\ y,\ z\right]$	`[1,28]`
2	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,31]`
3	$\left[y,\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[3,32]`
4	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ -z\right]$	`[4,25]`
5	$\left[\frac{3}{4},\ y,\ -z\right]$	`[5,30]`
6	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z\right]$	`[6,29]`
7	$\left[y,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[7,26]`
8	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[8,27]`
9	$\left[\frac{1}{4},\ -y,\ -z\right]$	`[9,20]`
10	$\left[y+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[10,23]`
11	$\left[-y,\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[11,24]`
12	$\left[\frac{1}{4},\ y+\frac{1}{2},\ z\right]$	`[12,17]`
13	$\left[\frac{1}{4},\ -y,\ z\right]$	`[13,22]`
14	$\left[\frac{1}{4},\ y+\frac{1}{2},\ -z\right]$	`[14,21]`
15	$\left[-y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[15,18]`
16	$\left[y+\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[16,19]`

Table 16: Wyckoff site: `32p`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	`[4]`
5	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	`[5]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[6]`
7	$\left[y,\ x,\ \frac{1}{2}-z\right]$	`[7]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[8]`
9	$[-x,\ -y,\ -z]$	`[9]`
10	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[10]`

continued ...

Table 16

No.	position	mapping
11	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11]
12	$\left[-x,\, y+\frac{1}{2},\, z\right]$	[12]
13	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[13]
14	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[17]
18	$\left[-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\, -x,\, z+\frac{1}{2}\right]$	[19]
20	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	[20]
21	$\left[-x,\, y+\frac{1}{2},\, -z\right]$	[21]
22	$[-x,\, -y,\, z]$	[22]
23	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[23]
24	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[25]
26	$\left[y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x,\, \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, y,\, z\right]$	[28]
29	$\left[x,\, \frac{1}{2}-y,\, z\right]$	[29]
30	$[x,\, y,\, -z]$	[30]
31	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[31]
32	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[32]